# Cole Snipes

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## SUMMARY

Full-stack software engineer and Data Analysis (B.S. Computer Science, May 2026) with production experience building AI-powered web applications, REST APIs, and machine learning pipelines. Proficient across the full development lifecycle : SvelteKit, React, Node.js, Firebase, and LLM API integration (Gemini, Claude, OpenAI). Delivered a capstone application serving a university physical therapy program through Agile sprints with a 5-person team. Seeking a full-stack developer or Data Analysis role to contribute immediately.

## EDUCATION

**Bachelor of Science in Computer Science** Graduated: May 2026

Major Concentration: Software Engineering · University of Indianapolis, Indianapolis, IN

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, TypeScript, C, C++, PHP, R, Rust, Swift, HTML, CSS, Sass/SCSS, Bash, Shell Scripting

**Frontend:** React, Svelte, SvelteKit, Next.js, Node.js, Vite, Tailwind CSS, Web Components, WebAssembly (WASM), Single Page Applications (SPA), Component-Driven Development, Responsive Web Design, Mobile-First Design, UI/UX Development, Server-Side Rendering (SSR), Static Site Generation (SSG), State Management, Cross-Browser Compatibility

**Backend:** Express.js, Spring Boot, Django, Flask, REST API Development, RESTful API Design, GraphQL, Third-Party API Integration, Webhook Integration, Serverless Functions, WebSockets, Authentication & Authorization (JWT, OAuth2), Session Management, CORS Handling, Rate Limiting, Error Handling & Logging, Performance Optimization

**AI / ML & LLMs:** Generative AI, Large Language Models (LLMs), Prompt Engineering, LLM API Integration (Claude, Gemini, OpenAI), Retrieval-Augmented Generation (RAG), AI Agents, Multi-Agent Systems, MCP (Model Context Protocol), Tool Use / Function Calling, AI Workflow Automation, Fine-Tuning, LangChain, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, OpenCV, Natural Language Processing (NLP), Supervised Learning, Unsupervised Learning, Neural Networks, Deep Learning, Predictive Modeling, Feature Engineering, Model Evaluation & Validation, Hyperparameter Tuning, A/B Testing

**Databases & Cloud:** Firebase (Firestore, Auth, Cloud Functions), Vercel, SQL, NoSQL, Data Pipelines, Data Cleaning & Preprocessing, Exploratory Data Analysis (EDA), Statistical Analysis

**DevOps & Tools:** Git, GitHub, GitHub Actions, Docker, CI/CD, Linux/Unix, macOS, SSH, VS Code, Jupyter, ESLint, Prettier, dotenv, npm/pnpm, Homebrew, Cron Jobs, Containerization, Package Management, Build Tools, Remote Development

**Testing:** Unit Testing, Integration Testing, End-to-End Testing (Playwright), TDD, API Testing, Selenium, Regression Testing, Load Testing, Mocking & Stubbing, Code Coverage, Debugging & Profiling, Static Analysis, Linting

**Practices:** Full-Stack Development, Agile/Scrum, Code Review, RBAC, JWT/OAuth2, REST API Design, Object-Oriented Programming (OOP), Functional Programming, Design Patterns, Event-Driven Architecture, Cross-Functional Collaboration, Technical Writing, Stakeholder Communication, Requirements Gathering

## PROJECTS

### Krannert School of Physical Therapy: Curriculum Web Application (SvelteKit 2, Firebase, Gemini AI)

- Built a full-stack AI web application that replaced a manual, spreadsheet-based CAPTE accreditation process, enabling DPT faculty to map courses and learning objectives against 93 CAPTE accreditation standards in a single interface.
- Architected the frontend in SvelteKit 2 + Svelte 5 and the backend in Firebase (Firestore, Auth, Cloud Functions), implementing RBAC across 4 user roles (student, faculty, admin, department chair) to enforce access at the API level.
- Integrated Gemini AI to surface curriculum alignment gaps, reducing cross-referencing time for administrators from hours to minutes per review cycle.
- Delivered across 5 Agile sprints with a 5-person cross-functional team; coordinated directly with the department chair to validate requirements against CAPTE review criteria.

### Predictive Analysis of Obesity Levels (Python, Keras, Scikit-learn, Pandas)

- Built an end-to-end ML pipeline (data cleaning, EDA, feature engineering, training, evaluation) on 2,111 patient records training and comparing a neural network (96% accuracy), KNN (95%), and random forest models on 16 features across 6 obesity classes.
- Identified key predictors via chi-square analysis ($\chi 2=448.9$, $p<0.001$) including family history, dietary habits, and physical activity frequency; visualizations reviewed and validated by faculty mentor Dr. Talaga.

### WebVPython WASM Runner: Glowscript Open Source (JavaScript, WebAssembly, postMessage API)

- Extended an open-source WASM runner to support interactive Python physics simulations fully in-browser, eliminating the requirement for a local Python runtime.
- Engineered secure cross-origin inter-frame messaging (postMessage API) and modified canvas rendering to enable screenshot export, expanding platform capabilities without breaking existing simulation behavior.

### n8n-to-Python Workflow Transpiler (Python, AST, CLI)

- Built a CLI tool that converts n8n visual automation workflows into standalone Python scripts via AST-based code generation, eliminating the n8n runtime dependency for production deployments.

- Supports core n8n node types including HTTP requests, data transformation, and conditionals, producing clean output targeted at DevOps engineers moving workflows to production.

## EXPERIENCE

**Cart Attendant** | Meijer, Indianapolis, IN January 2024 – Present

- Managed cart retrieval and floor safety across a high-volume store, maintaining consistent availability during peak hours with minimal supervision.
- Coordinated with store associates on spill response, cleanliness, and customer assistance, recognized for reliability and communication.

## AWARDS & RECOGNITION

- Recipient of the Richard Lugar Academic Scholarship
- Presented statistical research to Summer Research Institute faculty; selected for Scholar Showcase exhibition